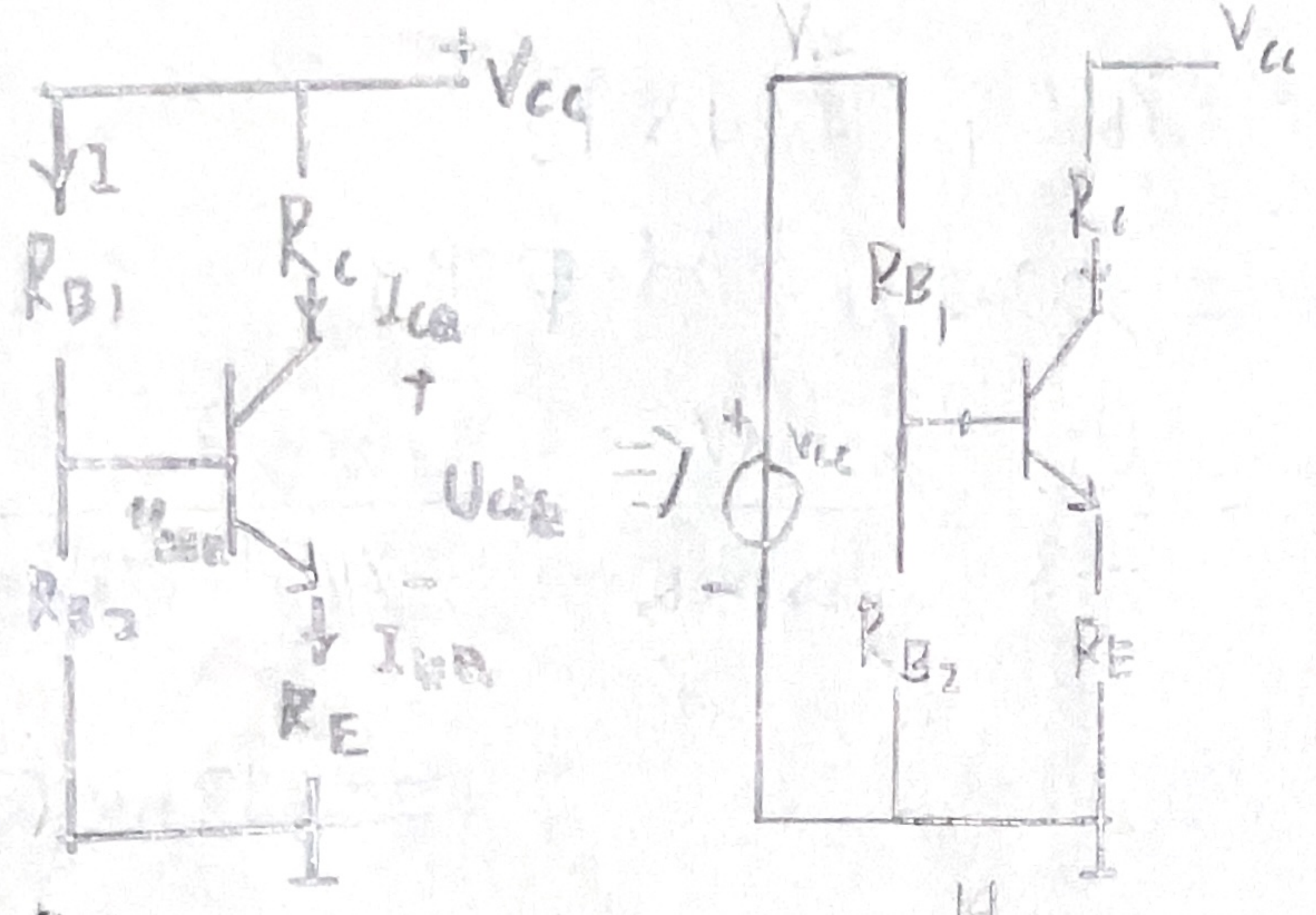
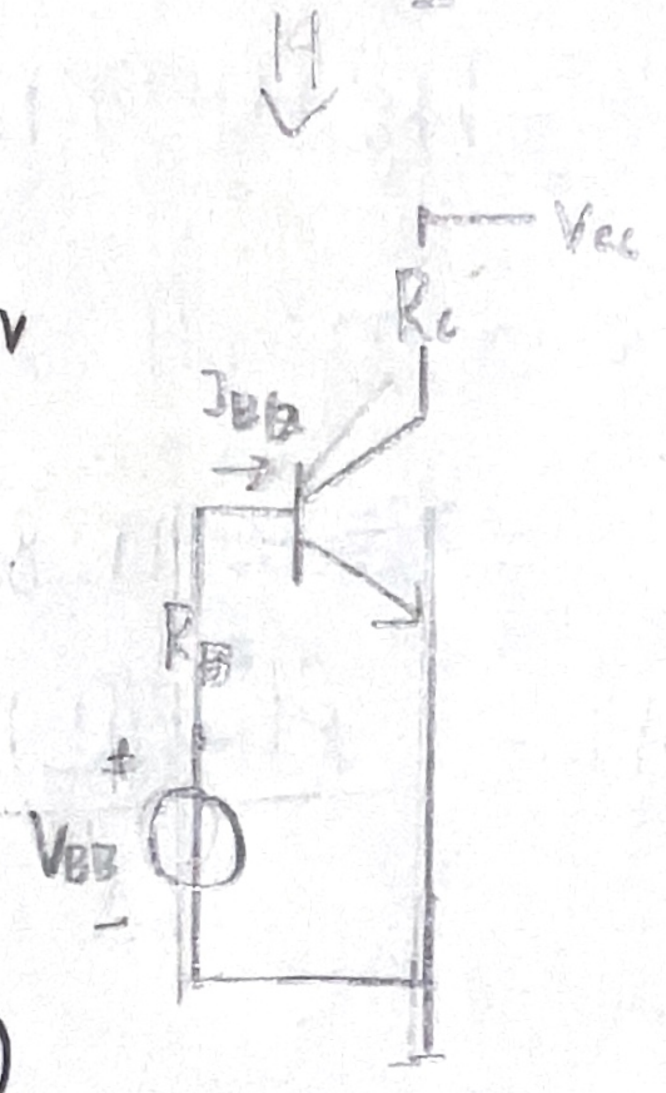
二、

共射

直流通路:





戴维南

$$V_{BB}=\frac{R_{B2}}{R_{B1}+R_{B2}}V_{cc}=\frac{10k\Omega}{39k\Omega+10k\Omega}\times 15V$$

$$=3.06V$$

输入电阻

(电压源短路,两端开路)

$$R_B=R_{B1}//R_{B2}$$

$$=7.96k\Omega$$

由KVL:

$$V_{BB}=I_{BQ}R_B+U_{BEQ}+I_{EQ}R_E$$

且有

$$I_{EQ}=(1+\bar{\beta})I_{BQ}$$

$$\therefore V_{BB}=I_{BQ}R_B+U_{BEQ}+(1+\bar{\beta})I_{BQ}R_E$$

$$\therefore I_{BQ}=\frac{V_{BB}-U_{BEQ}}{R_B+(1+\bar{\beta})R_E}$$

其中 $U_{BEQ}=0.7V$

$$\therefore I_{BQ}=\frac{3.06V-0.7V}{7.96k\Omega+101\times 1k\Omega}=0.216mA$$

$$\therefore I_{CQ}=\bar{\beta}I_{BQ}=2.17mA$$

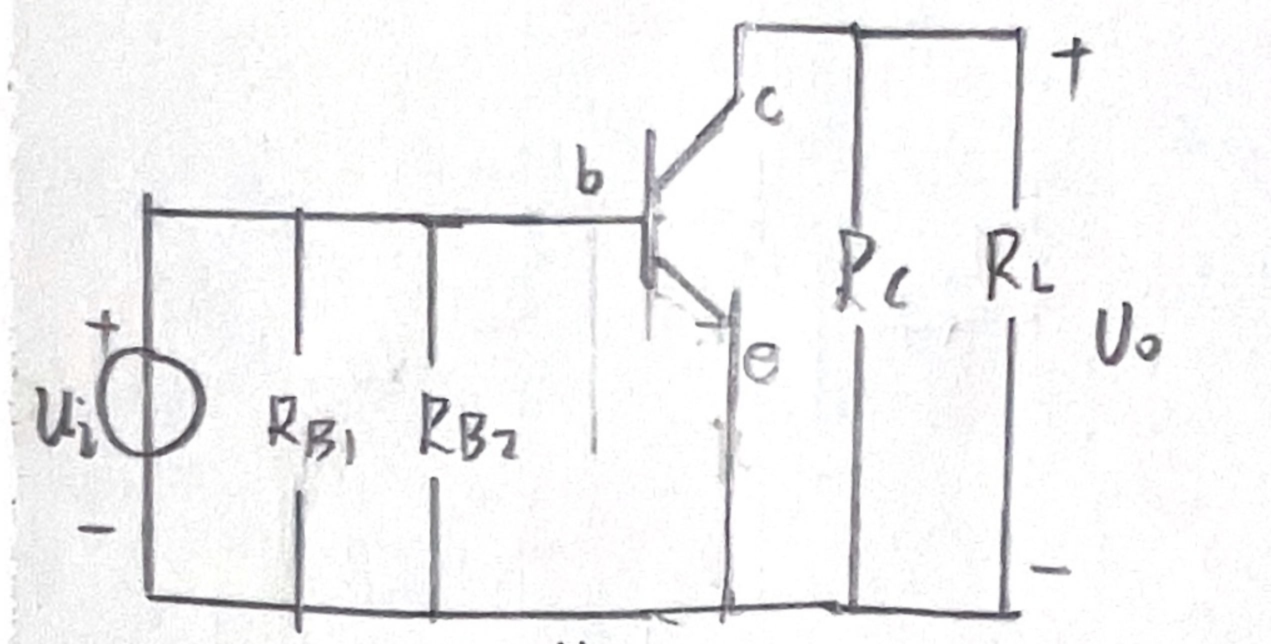
$$U_{CEQ}=V_{cc}-I_{CQ}\cdot R_c-I_{EQ}\cdot R_E$$

$$=V_{cc}-I_{CQ}(R_c+R_E)$$

$$=15V-2.17mA\times(2.7k\Omega+1k\Omega)$$

$$=\cancel{15V}-6.971V$$

交流通路



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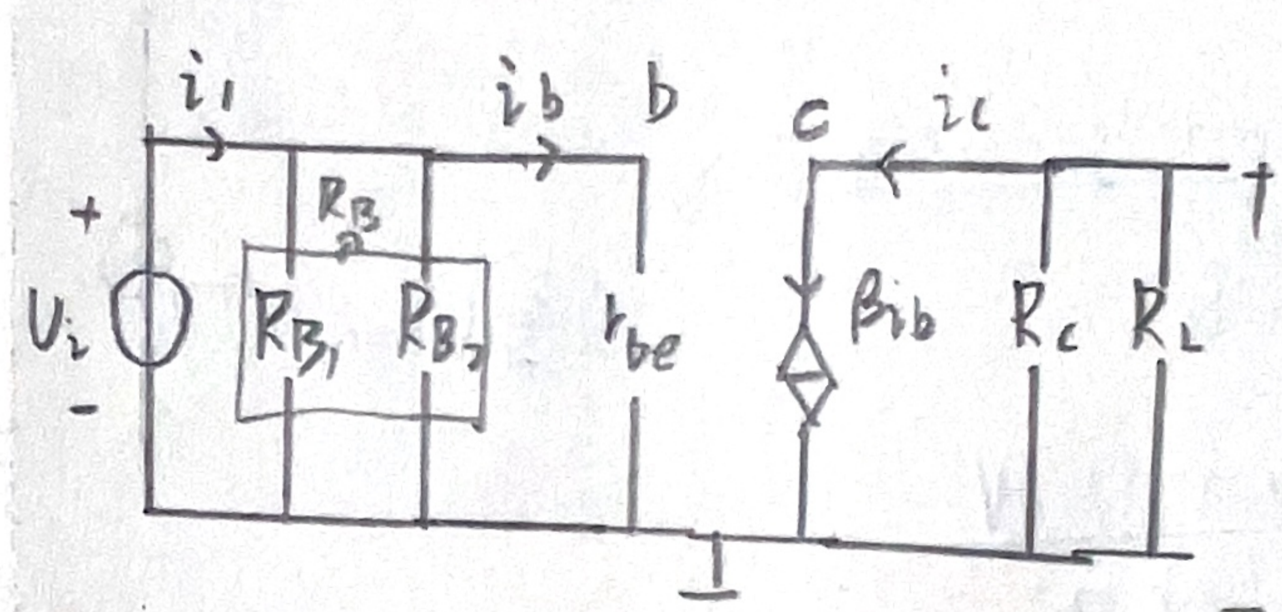

其中

$$r_{be}=r_{bb'}+(1+\beta)r_e$$

$$=r_{bb'}+(1+\beta)\frac{U_T}{I_{EQ}}$$

$$\cancel{=r_{bb'}+(1+\beta)}$$

$$=300\Omega+(1+100)\frac{26mV}{2.17mA+0.216mA}$$

$$=1.4k\Omega$$

$$\dot{A}_u=$$